

LINGUERE, Senegal

WMO: 616270

Lat: 15.383N

Lon: 15.117W

Elev: 69

StdP: 14.66

Time Zone: 0.00 (E00)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	60.2	62.6	22.5	17.0	81.8	25.6	19.6	82.5	14.2	80.0	12.4	80.5	3.4	40	0.382

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	29.8	109.9	72.6	107.8	72.5	105.8	72.5	82.7	94.4	81.8	92.8	81.0	91.5	4.6	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.0	156.1	87.2	79.1	151.5	86.4	78.6	148.7	85.9	46.4	95.6	45.5	93.1	44.7	91.6	91.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.1	10.0	9.0	DB	54.4	114.2	2.2	2.1	52.8	115.7	51.5	116.9	50.3	118.0	48.6	119.5	(4)
(5)			WB	44.5	85.9	2.9	2.4	42.4	87.7	40.7	89.1	39.1	90.5	37.0	92.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	85.6	77.5	81.2	86.0	88.6	90.9	90.6	87.6	85.3	85.1	88.5	85.5	79.6
	DBStd	5.52	4.24	4.57	4.84	4.73	3.96	3.20	3.02	2.80	2.73	2.79	3.08	4.08
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12977	851	874	1117	1159	1269	1218	1165	1093	1053	1194	1065	919
	CDD65	7502	386	454	652	709	804	768	700	628	603	729	615	454
	CDH74	101694	4227	5652	9252	10678	12287	11308	9571	7898	7419	10127	8185	5090
	CDH80	60859	2161	3226	5878	7198	8415	7309	5370	3833	3584	6115	5006	2764
Wind	WSAvg	4.4	4.6	5.0	4.6	5.0	4.9	5.2	5.0	3.8	3.3	3.3	3.5	4.4
Precipitation	PrecAvg	17.8	0.0	0.0	0.0	0.0	0.0	0.9	3.6	5.8	4.7	1.4	0.0	0.1
	PrecMax	30.8	1.4	1.3	0.6	0.2	0.6	5.2	10.9	15.3	14.3	8.4	0.7	1.2
	PrecMin	10.1	0.0	0.0	0.0	0.0	0.0	0.0	0.4	1.6	0.5	0.0	0.0	0.0
	PrecStd	5.0	0.2	0.2	0.1	0.0	0.1	0.9	2.3	2.6	2.7	1.7	0.2	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	99.7	105.0	109.2	112.2	112.7	111.3	104.2	100.4	100.1	106.7	105.2	102.2
		MCWB	65.0	66.2	70.4	73.9	73.4	74.6	76.9	78.0	80.4	74.4	72.8	68.7
	2%	DB	96.3	101.2	106.5	109.2	110.2	107.9	101.3	97.1	96.7	104.4	103.0	98.7
		MCWB	63.3	65.7	69.5	71.9	72.6	75.0	77.0	79.1	80.8	75.1	71.0	67.3
	5%	DB	93.5	98.2	104.0	106.7	107.7	105.5	98.9	94.9	94.7	102.5	101.3	96.3
		MCWB	62.6	64.9	69.3	71.0	73.1	75.4	77.4	79.2	80.8	74.8	70.1	66.5
	10%	DB	90.9	95.3	101.1	104.1	105.2	102.7	96.8	92.9	93.0	100.3	99.2	93.3
		MCWB	62.1	64.1	68.0	70.1	72.9	75.5	77.3	79.5	80.6	75.0	69.0	65.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.3	70.9	77.8	82.1	81.2	82.7	82.6	82.8	83.7	83.7	78.1	76.1
		MCDB	88.9	95.9	102.2	104.7	102.5	99.9	95.4	92.5	93.6	99.4	99.4	93.5
	2%	WB	67.1	68.8	74.4	78.3	78.3	80.4	80.7	81.7	82.6	81.8	75.8	72.3
		MCDB	88.9	94.2	99.8	100.1	101.8	97.3	93.2	91.3	92.3	95.2	95.4	91.7
	5%	WB	65.0	67.3	71.8	75.5	76.5	78.8	79.8	80.9	81.8	80.6	73.7	69.6
		MCDB	88.0	92.6	96.8	99.0	99.7	96.0	92.0	90.2	91.2	92.6	93.1	89.2
	10%	WB	63.4	65.8	69.5	73.0	75.2	77.5	79.0	80.3	81.2	79.6	71.9	67.6
		MCDB	87.1	90.8	94.9	96.7	97.8	94.9	90.9	89.2	90.1	90.6	93.0	86.8
Mean Daily Temperature Range	5% DB	MDBR	26.7	27.2	28.4	30.0	29.8	25.7	18.6	15.3	16.2	24.1	28.5	27.0
		MCDBR	30.6	30.2	31.7	32.1	31.3	29.1	21.9	18.2	18.6	27.6	31.2	31.7
		MCWBR	12.3	11.0	10.7	9.5	8.4	6.8	5.1	5.4	5.9	7.7	10.8	12.6
	5% WB	MCDBR	27.7	27.4	30.6	30.5	30.0	24.5	18.0	15.9	17.2	22.6	28.4	27.0
		MCWBR	11.8	10.7	12.3	12.0	9.5	8.0	5.9	5.9	6.5	7.1	12.1	12.8
Clear-Sky Solar Irradiance	taub	0.532	0.565	0.642	0.677	0.729	0.771	0.725	0.648	0.613	0.606	0.516	0.504	
	taud	1.797	1.728	1.585	1.568	1.478	1.476	1.587	1.747	1.812	1.796	1.916	1.904	
	Ebn at Noon	239	239	226	218	204	193	203	222	229	226	243	244	
	Edh at Noon	66	74	88	89	96	95	86	74	69	69	58	58	
All-Sky Solar Radiation	RadAvg	1638	1849	2114	2225	2197	2075	1939	1864	1838	1818	1657	1558	
	RadStd	92	96	76	73	77	50	51	56	69	66	80	61	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page